

Credit Card Default Prediction

Risk, interpretation and decision costs

Emerging Topics in FinTech

THE TEAM

Lei Peng

G2509090C · Part 1

Zhang Hanyu

G2509091L · Part 2

Zhou Xinzhe

G2509033F · Part 3

Xu Yiqun

G2509092H · Part 4

Contribution: 25% each

How we split and use the data

30,000 clients • 23 predictors •
22.12% default

Categories learned within each
training fold

Validation selects policies; historical
test reports outcomes

Partition and purpose

Fitting	18,000
----------------	--------

Five-fold CV and final fitting

Validation	6,000
-------------------	-------

Select decision thresholds

Test	6,000
-------------	-------

Evaluate frozen decisions

Source: data/splits; protocol_frozen.json; results/final/audit.json

Logistic regression: the linear reference

L2 model + fold-trained scaling + one-hot categories

12 declared settings; CV selected
C=0.1, unweighted

Original LR notebook was empty;
original process unverified

AP 0.4967

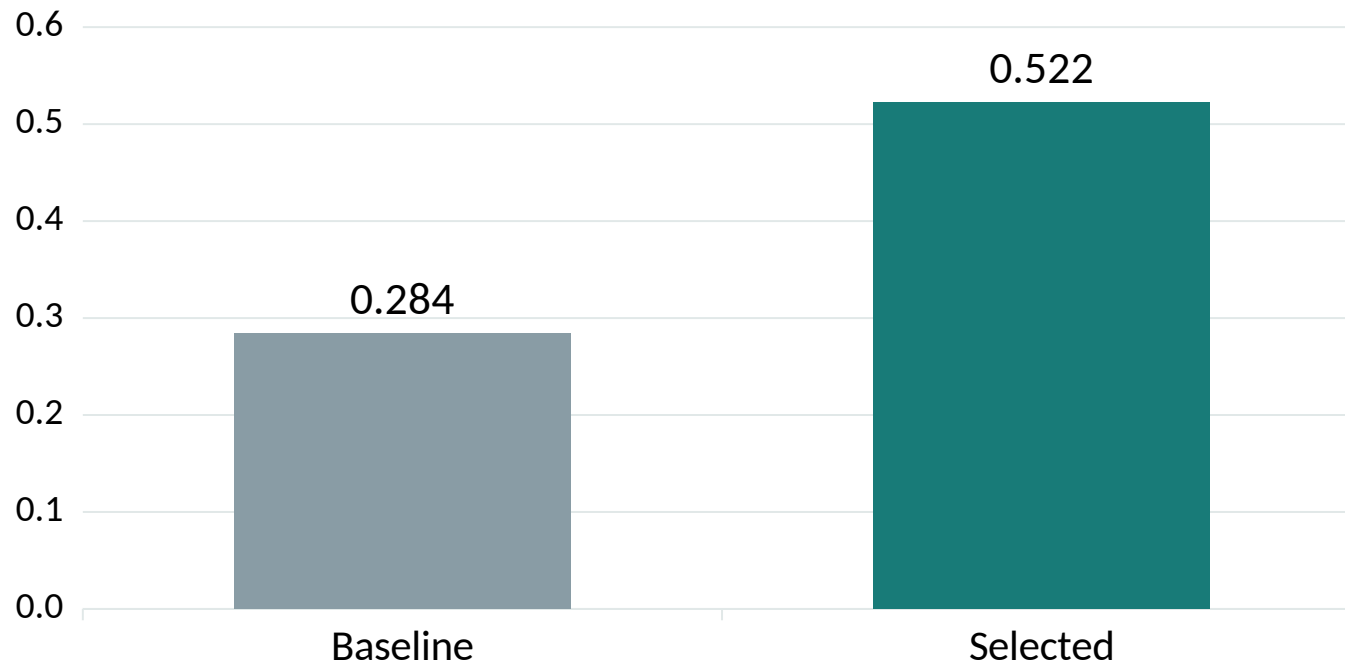
Baseline AP 0.4970

No ranking improvement from selection
LR* is a separate post-hoc supplement

Source: results/final/model_comparison.csv

Decision tree: controlling overfitting

Test average precision



21 configurations: depth × minimum leaf size

Unrestricted baseline fits training folds almost perfectly

Selected minimum leaf size: 100 observations

A large baseline gap signals overfitting.
CV selection improves this tree's ranking.

Source: results/final/model_comparison.csv

Reading the tree's first decision

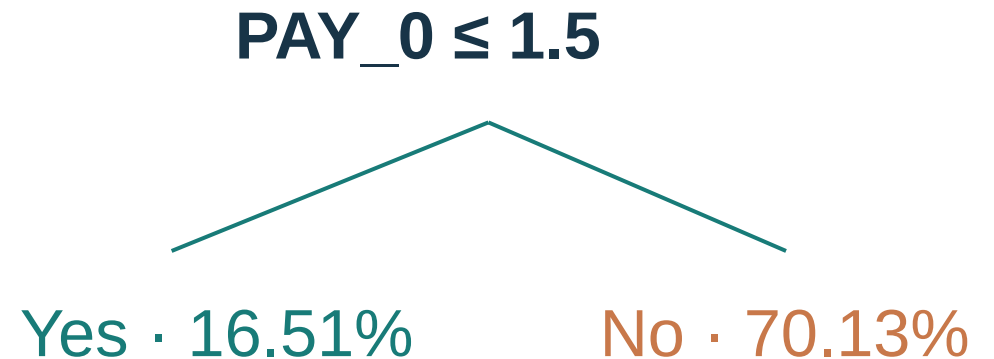
Root rule: recent repayment status
 $\text{PAY_0} \leq 1.5$

Selected tree: depth 18, with 134
leaves

Node proportions describe this fitting
sample

Source: saved decision-tree model; integration audit/report evidence.

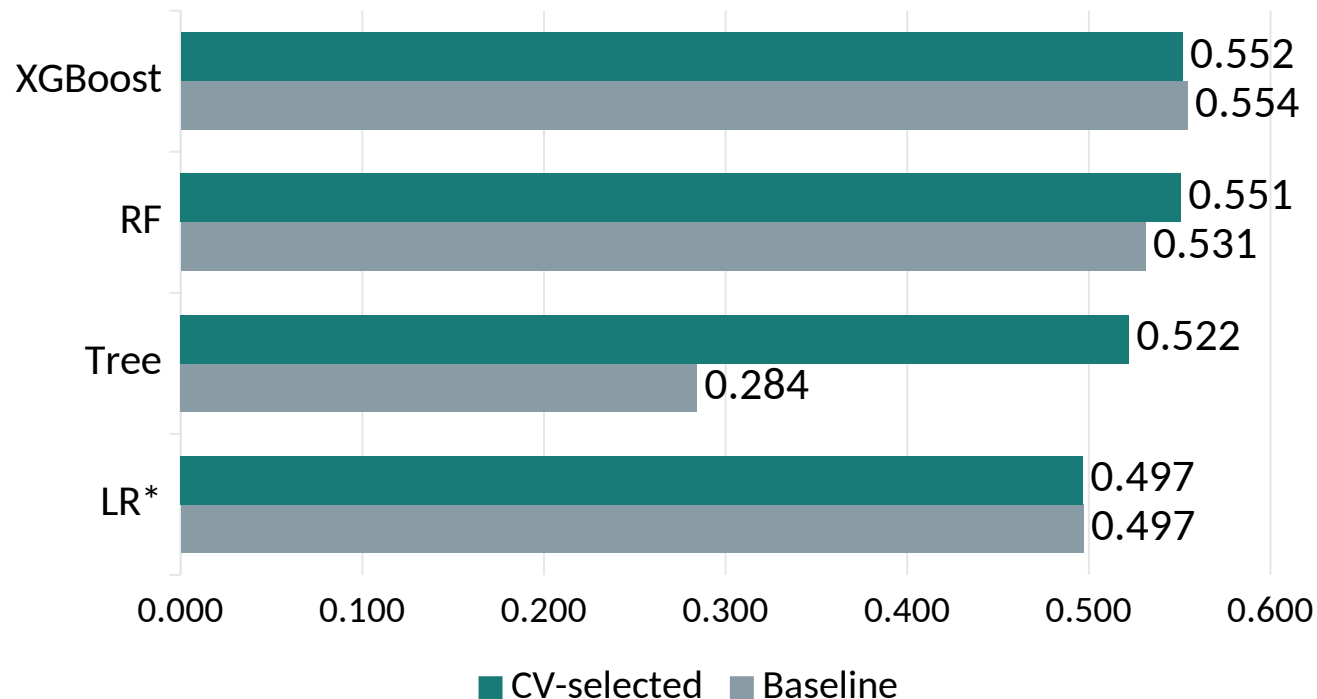
Recent repayment status



Default proportion in each fitting-sample branch.
These are associations, not causal effects.

Ranking performance across four models

Test average precision



Decision tree gains most on this split

RF and XGBoost have the highest selected AP

CV selection precedes test comparison

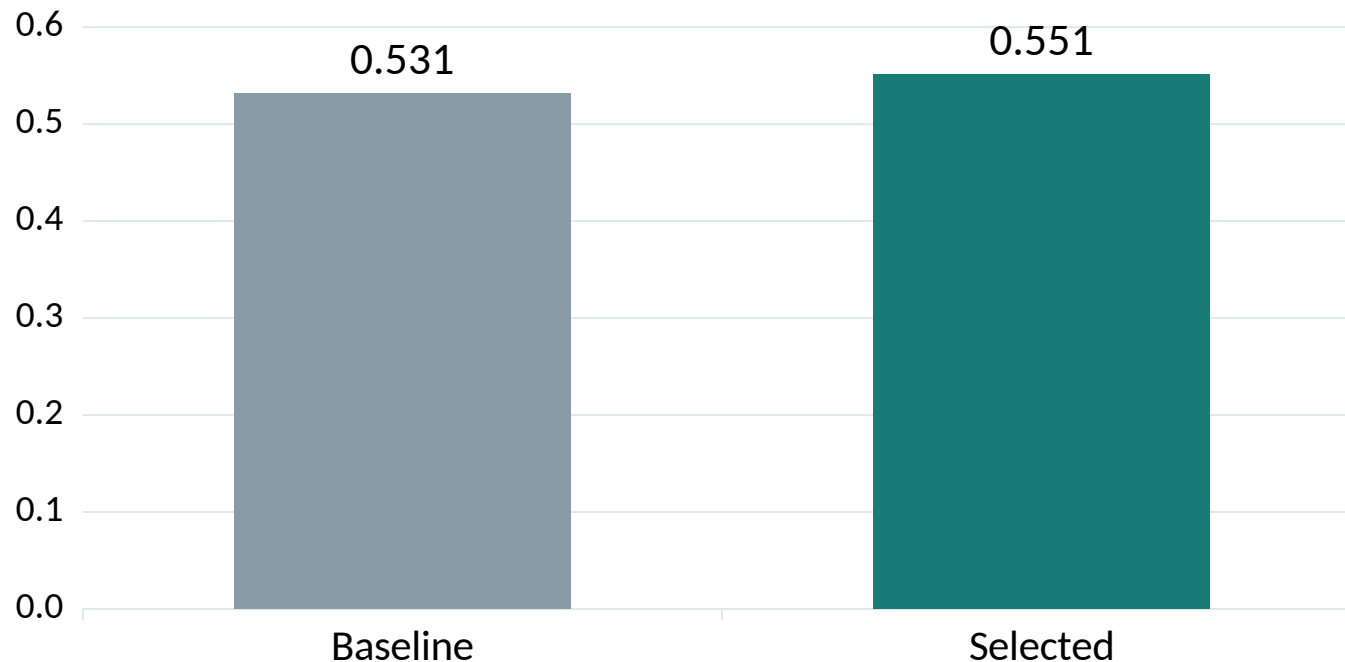
Baseline and selected models are both retained.

Small ensemble differences do not establish superiority.

Source: results/final/model_comparison.csv

Random forest: averaging and regularization

Test average precision



Baseline + 23 seeded candidate settings

Selected: 500 trees, depth 8, minimum leaf 2

Positive-class weight 3; feature fraction 0.5

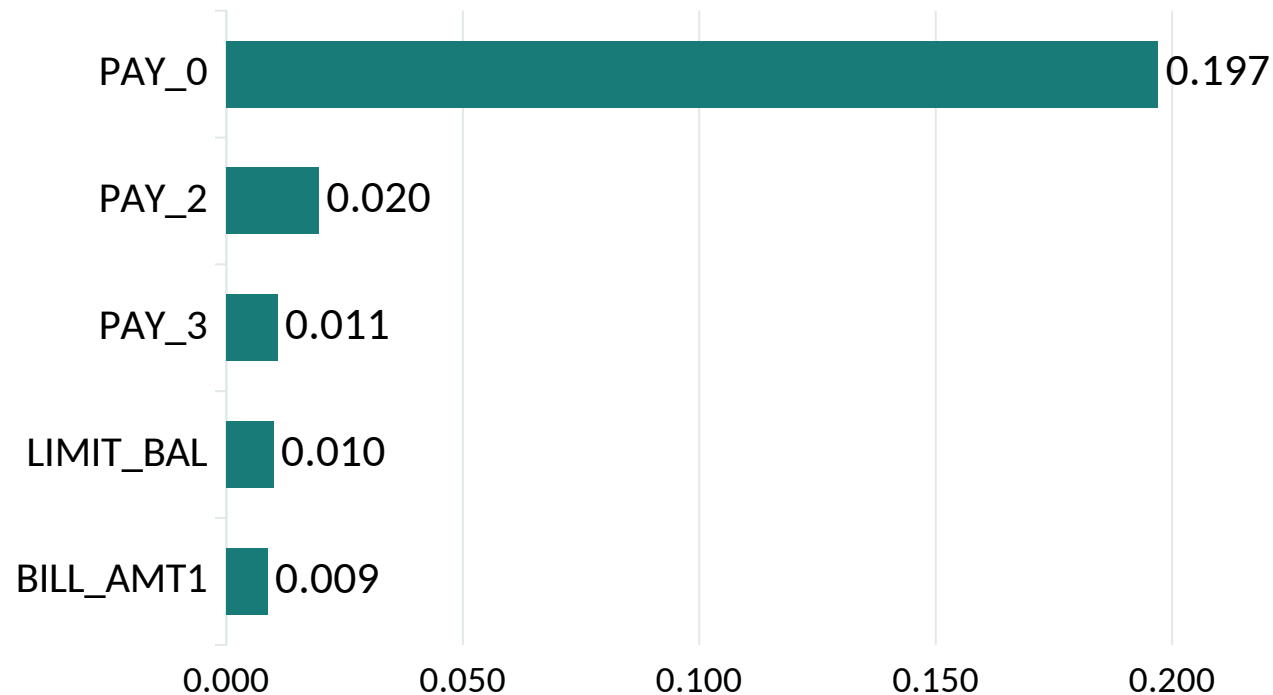
Averaging reduces dependence on one tree.

Weights also change the score distribution.

Source: results/final/model_comparison.csv

Random forest: which features matter?

Mean decrease in validation AP



PAY_0 has the largest validation AP decrease

Permutation measures predictive reliance

Correlated variables can substitute for each other

Five saved shuffles per feature.

Predictive association, not causality.

Source: results/rf/validation_importance.csv

Random forest: two illustrative errors

At 0.5: 637 missed defaults; 573 false alarms

Missed default · row 982

Score 0.1012; actual default = 1

False alarm · row 28747

Score 0.9415; actual default = 0

Source: results/rf/misclassification_cases_05.csv

Why the predictions can fail

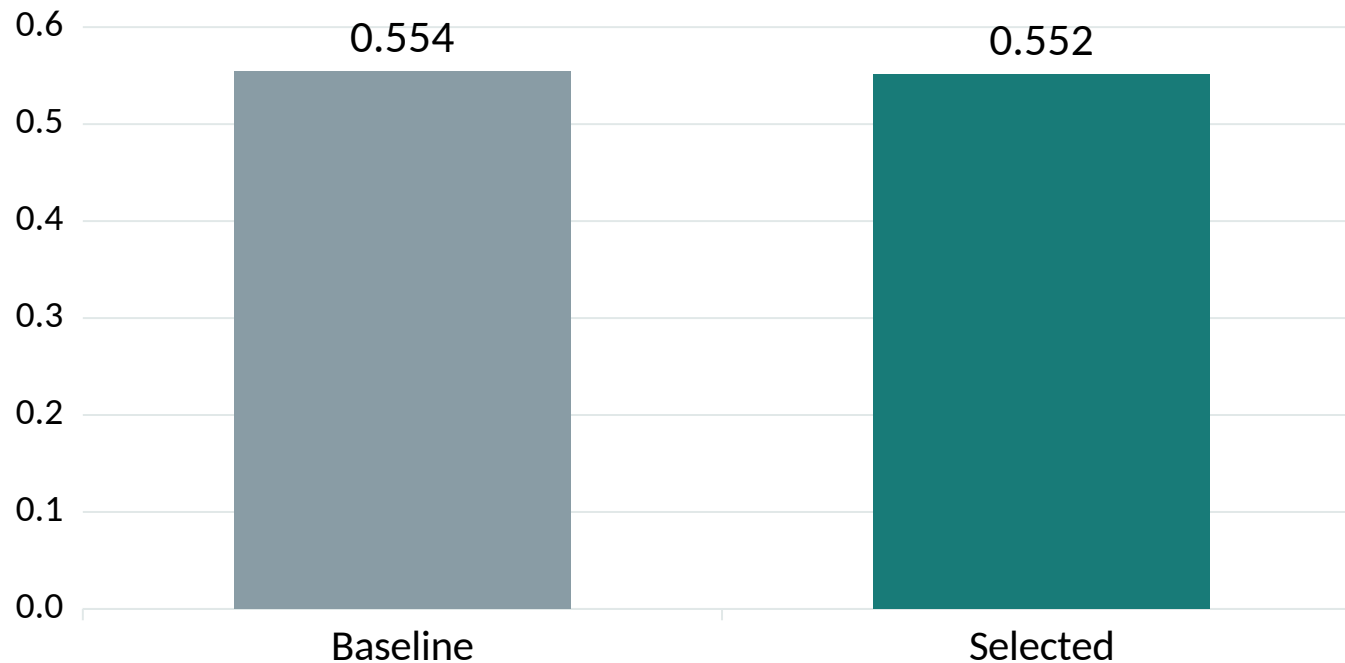
Missed default: PAY_0 = -2

False alarm: PAY_0 = 3

Illustrative extreme errors, not typical clients.

XGBoost: no gain in test ranking

Test average precision



**24 configurations; select by
CV AP**

Selected: 150 trees, depth 4,
rate 0.03

Selected ROC-AUC: 0.7762

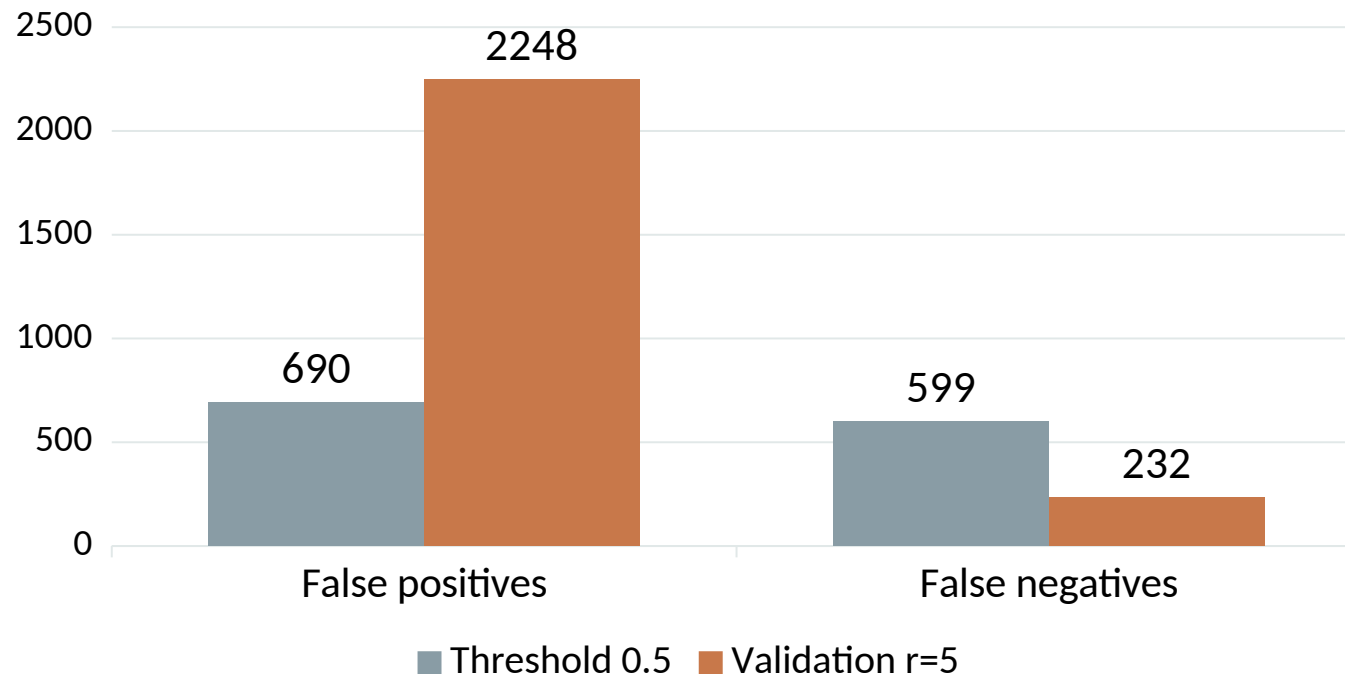
Tuned – baseline AP: -0.0028

Conditional 95% interval: [-0.0106,
0.0054]

Source: results/final/model_comparison.csv

XGBoost: fewer misses, more reviews

Test confusion trade-off



Validation $r=5$ cutoff: 0.3150

Recall 54.9% $\rightarrow$ 82.5%; alerts 55.7%

Cost 3,685 $\rightarrow$ 3,408: 7.5% reduction

Same ranking scores, different action policy.

Review costs are hypothetical; capacity is unmeasured.

Source: results/final/model_comparison.csv

What the results mean for credit review

Ranking, explanations, and workload need separate evidence

No clear ensemble winner from these small differences

Next: fresh temporal data, calibration, subgroup errors

Discussion

Set realistic costs and review capacity.
Validate the policy on fresh data.

Historical educational study; no causal or deployment-readiness claim.

Backup A • AP, ROC-AUC and threshold metrics

AP = sum of recall increments ×
precision

ROC-AUC measures
positive/negative score ordering

Precision, recall, F1 and accuracy
require a threshold

Two different definitions

Saved continuous scores supply ranking
metrics.

Changing the cutoff changes decisions
only.

Source: [scikit-learn average_precision_score documentation](#); part4/evaluation.py

Backup B • Cost ratios and fixed-prediction uncertainty

Predeclared ratios: $r = 1, 3, 5, 10$

Validation minimum cost; ties prefer largest cutoff

1,000 stratified paired bootstrap resamples

What the intervals cover

Fixed models / thresholds / class counts

Excludes fitting / selection uncertainty

Excludes population / prevalence shifts

Source: frozen protocols; integration/uncertainty.py; bootstrap_intervals.csv

Backup C • Data provenance and evidence boundaries

UCI cleaning reproduced exactly from downloaded source

EDUCATION 0/5/6 → 4; MARRIAGE
0 → 3

Retained duplicate vectors can span partitions

Sources: results/final/source_audit.json; integration/LOGISTIC_PROVENANCE.md

The limits of historical evidence

Original LR process remains unverified.

LR* source was committed before its one run.

No new unseen-test claim is made.

Backup D • Sources, responsibilities and reproducibility

UCI: doi.org/10.24432/C55S3H

scikit-learn: [average_precision_score](https://scikit-learn.org/stable/modules/average_precision_score.html)
documentation

XGBoost paper:
doi.org/10.1145/2939672.2939785

Member contributions • 25% each

Lei Peng • G2509090C

Part 1 — Data and logistic regression

Zhang Hanyu • G2509091L

Part 2 — Decision tree and overfitting

Zhou Xinzhe • G2509033F

Part 3 — Random forest and error analysis

Xu Yiqun • G2509092H

Part 4 — XGBoost and threshold trade-offs

Sources: Group 5 repository, frozen protocols, row-level predictions and integration audit.